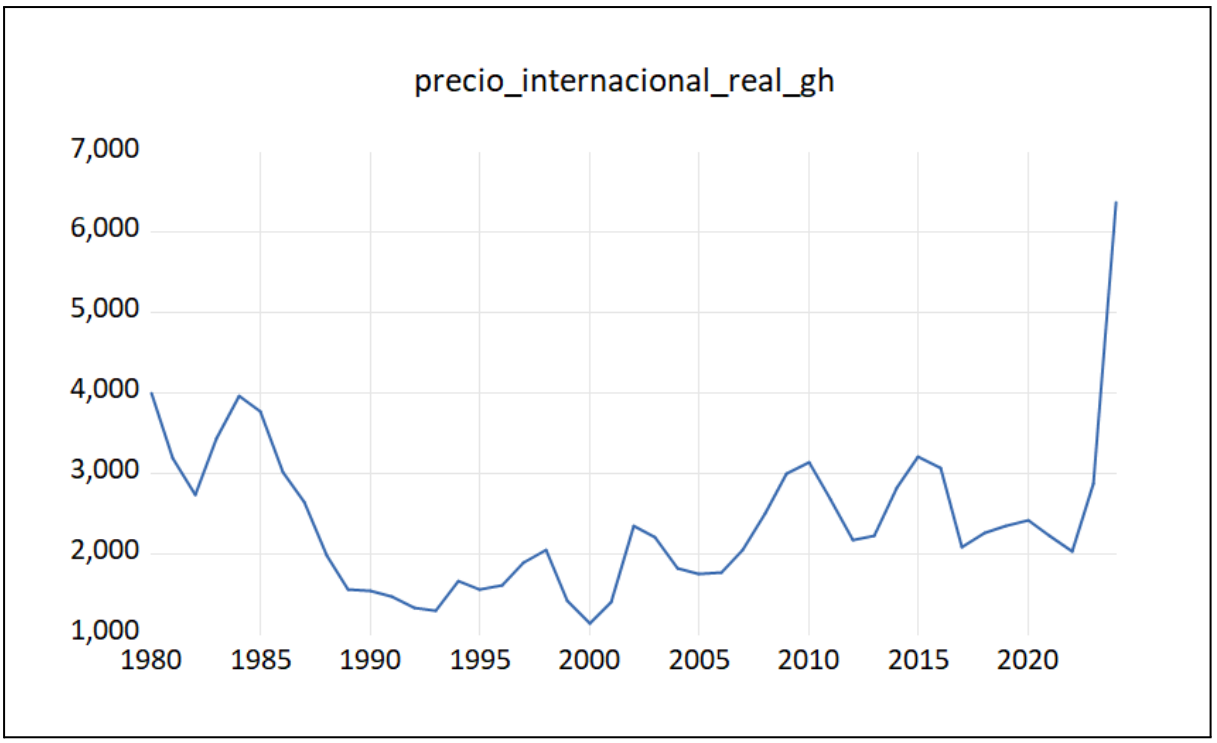


En el gráfico se observa una fuerte caída entre 1985 y 1990, luego cierta estabilidad, y finalmente una gran subida en los últimos años del precio internacional del cacao.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-1.243925	0.8882
Test critical values:	1% level	-4.186481	
	5% level	-3.518090	
	10% level	-3.189732	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:49
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	-0.172549	0.138713	-1.243925	0.2210
D(PRECIO INTERNACIONAL REAL GH	0.616396	0.226211	2.724873	0.0096
C	221.7287	385.0076	0.575907	0.5680
@TREND("1980")	11.30600	7.774911	1.454164	0.1539

R-squared	0.240057	Mean dependent var	74.03259
Adjusted R-squared	0.181600	S.D. dependent var	682.1569
S.E. of regression	617.1163	Akaike info criterion	15.77640
Sum squared resid	14852469	Schwarz criterion	15.94023
Log likelihood	-335.1926	Hannan-Quinn criter.	15.83682
F-statistic	4.106556	Durbin-Watson stat	1.119953
Prob(F-statistic)	0.012631		

Como la tendencia determinística no resulta significativa (p-valor = 0.1539), vuelvo a hacer el test con intercepto.

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: Constant
Lag Length: 2 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-0.566130	0.8673
Test critical values:	1% level	-3.596616	
	5% level	-2.933158	
	10% level	-2.604867	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:49
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	-0.081703	0.144319	-0.566130	0.5746
D(PRECIO INTERNACIONAL REAL GH	0.834834	0.227768	3.665281	0.0008
D(PRECIO INTERNACIONAL REAL GH	-0.628098	0.235939	-2.662122	0.0113
C	247.0291	340.6667	0.725134	0.4728

R-squared	0.316966	Mean dependent var	86.39314
Adjusted R-squared	0.263043	S.D. dependent var	685.5347
S.E. of regression	588.5057	Akaike info criterion	15.68344
Sum squared resid	13160881	Schwarz criterion	15.84894
Log likelihood	-325.3523	Hannan-Quinn criter.	15.74410
F-statistic	5.878048	Durbin-Watson stat	1.412468
Prob(F-statistic)	0.002123		

Nuevamente, el intercepto no resulta significativo (p-valor = 0.4728), por lo que realizo el test sin tendencia ni intercepto.

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: None
Lag Length: 2 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		0.496799	0.8186
Test critical values:	1% level	-2.621185	
	5% level	-1.948886	
	10% level	-1.611932	

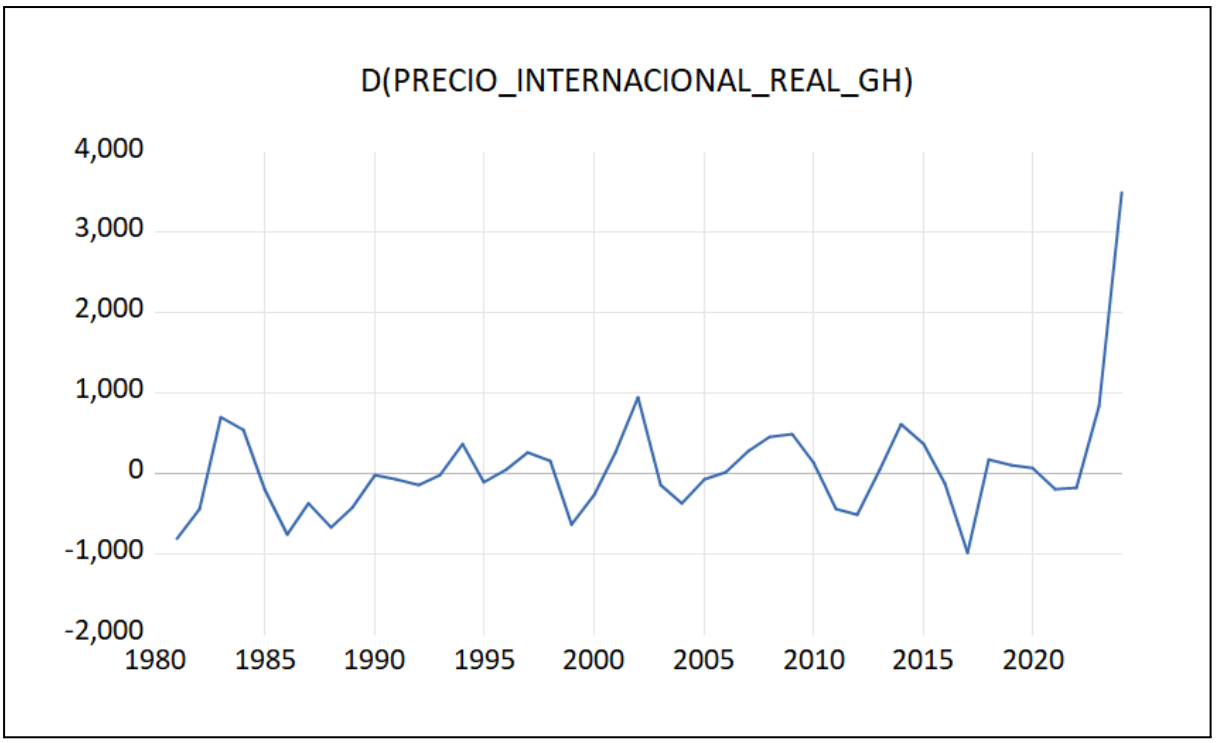
*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:50
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	0.019113	0.038472	0.496799	0.6221
D(PRECIO INTERNACIONAL REAL GH	0.802145	0.221901	3.614878	0.0008
D(PRECIO INTERNACIONAL REAL GH	-0.682925	0.222133	-3.074394	0.0038

R-squared	0.307515	Mean dependent var	86.39314
Adjusted R-squared	0.272003	S.D. dependent var	685.5347
S.E. of regression	584.9171	Akaike info criterion	15.64957
Sum squared resid	13342993	Schwarz criterion	15.77369
Log likelihood	-325.6409	Hannan-Quinn criter.	15.69506
Durbin-Watson stat	1.441096		

Como el p-valor asociado al test estadístico es 0.8186 no rechazo la hipótesis nula de raíz unitaria por lo que debo diferenciar la serie:



Notar la presencia de un valor atípico en los últimos años (2023-2024) donde el precio internacional se dispara.

Realizo el test de Dickey Fuller nuevamente para asegurarme de que no haya que diferenciar nuevamente la serie:

Null Hypothesis: D(PRECIO_INTERNACIONAL_REAL_GH) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.690253	0.0341
Test critical values:	1% level	-4.192337	
	5% level	-3.520787	
	10% level	-3.191277	

*MacKinnon (1996) one-sided p-values.

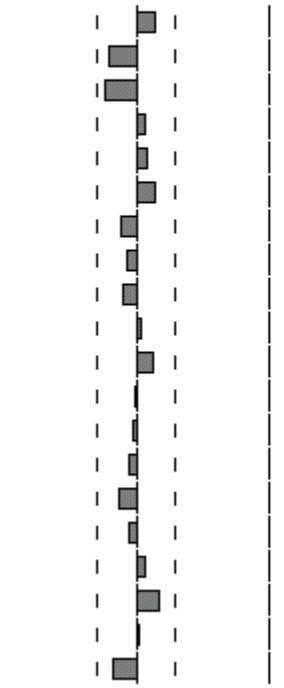
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO_INTERNACIONAL_REAL_GH,2)
Method: Least Squares
Date: 10/20/26 Time: 17:58
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(PRECIO_INTERNACIONAL_REAL_GH(-1	-0.959795	0.260089	-3.690253	0.0007
D(PRECIO_INTERNACIONAL_REAL_GH(-1	0.716317	0.215317	3.326810	0.0020
C	-293.3396	192.6193	-1.522898	0.1361
@TREND("1980")	14.97777	7.259097	2.063311	0.0460

R-squared	0.314819	Mean dependent var	93.76005
Adjusted R-squared	0.260726	S.D. dependent var	651.7977
S.E. of regression	560.4226	Akaike info criterion	15.58565
Sum squared resid	11934794	Schwarz criterion	15.75114
Log likelihood	-323.2987	Hannan-Quinn criter.	15.64631
F-statistic	5.819927	Durbin-Watson stat	1.563827
Prob(F-statistic)	0.002247		

Como la tendencia resulta significativa y el estadístico del test tiene un p-valor = 0.0341, rechazo la hipótesis nula de raíz unitaria en la serie diferenciada, al 5% de significatividad.

Date: 10/14/26 Time: 18:52
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.145	0.145	0.9644	0.326
		2	-0.219	-0.245	3.2214	0.200
		3	-0.245	-0.185	6.1186	0.106
		4	0.072	0.096	6.3733	0.173
		5	0.084	-0.035	6.7320	0.241
		6	0.148	0.144	7.8796	0.247
		7	-0.110	-0.122	8.5321	0.288
		8	-0.079	0.013	8.8812	0.352
		9	-0.099	-0.091	9.4395	0.398
		10	0.037	-0.017	9.5193	0.484
		11	0.133	0.120	10.582	0.479
		12	-0.000	-0.103	10.582	0.565
		13	-0.017	0.120	10.601	0.644
		14	-0.056	-0.072	10.807	0.701
		15	-0.137	-0.147	12.110	0.671
		16	-0.054	-0.016	12.319	0.722
		17	0.082	-0.035	12.816	0.748
		18	0.181	0.197	15.354	0.638
		19	0.019	-0.059	15.385	0.698
		20	-0.186	-0.075	18.284	0.569

El correlograma de la serie diferenciada se comporta como ruido blanco, lo cual tiene sentido teniendo en cuenta que se trata de la serie de precio internacional del cacao, que al ser una commodity es normal que se comporte como un *random walk*.